

Recovering Motion-Referring Expressions in Moving-Camera Videos via Global Motion Compensation

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ABSTRACT

Referring multi-object tracking takes a video and a natural-language expression and tracks every object that the expression describes, and many expressions refer to how an object moves rather than how it looks. Hosts that answer such queries read motion from raw bounding-box displacement, which a moving camera corrupts: a parked car sweeps across the frame and is scored as “moving,” while a car keeping pace with traffic reads as static and is dropped. An emerging motion-language line fuses object dynamics with text yet leaves the camera in the signal, while a mature ego-motion-compensation line removes the camera but never reaches language; the two have not met. We fill that intersection with a plug-in module at the decision level: per-frame homographies compose into a cumulative transform, residual velocity is read at several temporal gaps, a two-tower encoder aligns it with the expression, and the score is added to the host logit without host retraining. Aligning ego-compensated motion with the expression recovers the moving class that camera-naïve hosts miss: on iKUN it lifts moving-class HOTA by +8.83 (native 20.25 $\rightarrow$ 29.08, $n=5$), and an ablation identifies ego-compensation as the only engineered component with a significant effect on that alignment (-1.77 , $p<0.01$). Across the three host-split settings we study (two architectures, iKUN and FlexHook, the latter on both Refer-KITTI splits), the recovered signal is monotone-inverse to a host’s native motion ability. In pooled HOTA the module matches iKUN’s published score and improves FlexHook’s published V2-split score outright (+0.281), while on the V1 split it trails FlexHook’s published number under a known reproduction gap.

KEYWORDS

referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

1 INTRODUCTION

Referring multi-object tracking (RMOT) takes a video and a free-form sentence and outputs the trajectories of every object that the sentence describes; many of those sentences refer to how an object moves rather than to how it looks. A query for “moving cars” or “a vehicle turning left” asks a tracker to reason about motion, yet the hosts that answer such queries read motion straight from raw bounding-box displacement. That shortcut breaks the moment the camera itself moves. When the platform drives forward or pans, every box in the frame shifts, so a parked car accrues a large image-plane displacement that has nothing to do with the car. The host cannot tell the camera’s motion from the object’s, and the failure is systematic in both directions: a stationary car drifting across the

frame is scored as “moving”, while a car moving with the traffic is read as nearly static and dropped from the result. The signal a motion-class query depends on is exactly the signal a moving camera corrupts.

This failure persists because the two research lines that could fix it have not met. Motion-language RMOT does fuse object dynamics with text. VMRMOT [9] encodes position, direction, distance-trend, and speed-trend descriptors and aligns them with the expression, but it takes those descriptors directly from image-plane box trajectories, so its motion channel inherits the ego-motion contamination intact: a parked car still carries a nonzero speed-trend whenever the camera moves. The RMOT lineage it builds on [2, 6, 14] shares the same camera-naïve reading of motion. Plain multi-object tracking, meanwhile, has solved camera compensation. BoT-SORT [1] and UCMCTrack [15] estimate the inter-frame camera transform and warp predictions back onto a stable frame, but they spend the corrected signal entirely on association gating and never expose it to language. One line fuses motion with text but leaves the camera in it; the other removes the camera but never speaks to text. No prior RMOT aligns ego-compensated motion with the referring expression; platform pose is in principle available from KITTI’s GPS/IMU stream or a ground-plane projection, but no RMOT method routes it to language, and a feature-based estimate keeps the module sensor-free and host-agnostic.

We occupy that intersection with a plug-in module that operates at the decision level. The module composes per-frame homographies into a cumulative transform, warps the original object centroids through that chain, and reads residual velocity at several temporal gaps, yielding a motion descriptor that reflects the object rather than the camera and persists across the window a motion verb spans. A lean two-tower encoder aligns this descriptor with the expression, and the alignment score is added to the host logit, leaving the host’s appearance grounding untouched. The design rests on a falsifiable claim: ego-compensation should contribute over and above merely having a motion descriptor. Replacing residual velocity with raw, uncompensated velocity should then cost moving-class recovery, and Section 4 shows that it does — feeding raw velocity costs moving-class recovery, the largest component effect — while the bulk of the recovery comes from aligning motion with language at all.

This work connects the mature ego-motion-compensation machinery of plain tracking with the emerging motion-language line of RMOT, and makes the following contributions.

- A decision-level ego-motion-compensation module that attaches to an existing RMOT host with no retraining of the host and no detector swap, requiring only a six-coefficient per-host fusion table rather than a learned fusion head.
- A cross-architecture study over three host-split settings — iKUN on Refer-KITTI V1, FlexHook on V1 and V2, all

multi-seed — showing that the module’s gain forms a monotone-inverse ordering against a host’s native motion ability; the two lower-deficit FlexHook settings gain little, consistent with the characterization, though as one architecture on two splits they are not independent confirmations.

- Ablations ($n=5$) that credit motion–language alignment as the lever recovering the moving class (+8.83 over the native host), with ego-compensation the only engineered component whose removal costs a significant amount (a further +1.77, Welch $p=0.006$), and decision-level fusion as the viable injection site, after feature-level injection of motion into the visual encoder regresses sharply.

On iKUN [2], the module matches the published pooled HOTA [8] while lifting the motion class that the pooled number, dominated by a large appearance majority, otherwise hides; on FlexHook [6], evaluated on the Refer-KITTI V2 split, it improves the pooled score outright. These gains hold at a fixed tracker and are orthogonal to detector quality: reaching the top of the leaderboard requires a stronger tracker than the public detections expose, which is out of scope for a plug-in that changes neither the detector nor the host.

2 RELATED WORK

2.1 Referring MOT and motion-language fusion

Referring multi-object tracking (RMOT) extends multi-object tracking with a free-form expression, tracking only the objects that the expression describes. TransRMOT [14] established the task on Refer-KITTI with an end-to-end transformer that grounds language against per-frame queries, and iKUN [2] showed that a knowledge-unification module can speak to a frozen tracker without retraining, decoupling grounding from association. FlexHook [6] revisits the two-stage referring-by-tracking design and strengthens the cross-modal hook between detections and text. A broader RMOT lineage has since pushed grounding quality on Refer-KITTI through multilevel semantic interaction (MLS-Track [10]), deep cross-modality fusion (DeepRMOT [4]), cognitive disentanglement of what-versus-where cues (CDRMOT [7]), and robust language guidance [5], yet each still reads motion, where it uses motion at all, from raw image-plane geometry. These methods grew out of appearance grounding, and the descriptors they key on are dominated by color, category, and position. Vision-language matching for referring tasks inherits this bias from its CLIP-style backbones, which were trained to associate static images with text and carry no notion of how an object moves. Motion-class expressions such as “moving”, “turning”, and “parked” are therefore served only weakly, if at all, even though they hinge on a property no appearance feature encodes. VMOT [9] makes the motion channel explicit: it encodes position, direction, distance-trend, and speed-trend descriptors, then fuses them with the expression inside a multi-modal large language model, and it currently leads motion-language RMOT on Refer-KITTI (53.00 HOTA on V1, 35.21 on V2). The descriptors are taken directly from image-plane box trajectories, so a parked car carries a nonzero speed-trend whenever the camera moves and a static scene acquires a spurious direction. None of these hosts model camera ego-motion, which leaves the inferred displacement entangled with the platform’s own translation, exactly the signal a motion-class query

depends on. TempRMOT [16] sits apart: its recurrent memory already aggregates trajectory history across frames, so an external motion module is redundant rather than complementary, and we treat hosts with native temporal memory as out of scope.

2.2 Camera and ego-motion compensation in tracking

Compensating for a moving camera is a well-studied component of plain multi-object tracking, where it stabilizes the predicted box before data association. BoT-SORT [1] adds a global motion compensation step that warps Kalman predictions by an estimated inter-frame transform, recovering the association accuracy lost when the platform pans or drives. UCMCTrack [15] pushes the same idea onto the ground plane, applying a uniform camera-motion model so that localization survives sustained ego-motion without re-tuning per sequence. BoT-SORT rests on the classical feature-and-fit pipeline that estimates an inter-frame homography from sparse correspondences: ORB keypoints [12] matched across frames, with RANSAC [3] rejecting outliers before the transform is fit. UCMCTrack instead avoids per-frame feature matching, applying a uniform sequence-level camera-motion model on the ground plane. What the two share is the point worth drawing out: neither composes the correction into a persistent, ego-stable trajectory over the temporal window a motion verb ranges over; the corrected estimate is consumed immediately by association and discarded. Compensation feeds the geometric gate that decides which detection continues which track, measured by tracking metrics such as HOTA [8]. Because the goal is association rather than description, the compensated displacement is never named as a feature, never accumulated into a stable velocity, and never exposed to a language model. These trackers therefore stay blind to what the objects are being asked to do, even as they hold the very signal a motion-class query needs.

2.3 The unfilled intersection

Two mature lines run in parallel without meeting. Motion-language RMOT fuses object dynamics with text but reads those dynamics from camera-contaminated box displacement, so its motion channel degrades exactly when the platform moves and the speed and direction it reports drift away from what the object is physically doing. Ego-motion compensation removes that contamination, yet it compensates only frame to frame and spends the corrected signal entirely on association gating, with no path to the referring expression. Neither line, on its own, gives a language model a motion descriptor that is both ego-stable and accumulated over the window a motion verb spans. We compose the per-frame homographies into a cumulative transform, warp the original object centroids through that chain, and read residual velocity at several temporal gaps, so the motion handed to the aligner reflects the object rather than the camera and persists across the frames the expression refers to. This ego-compensated, multi-frame motion is then aligned with the language embedding and fused with the host score at decision level, which keeps the host tracker untouched and lets the same module attach to more than one architecture. Decision-level fusion is deliberate: it adds the motion evidence after the host has scored appearance, so the appearance grounding the host already does well

is preserved and only the motion-class deficit is supplied. This intersection of ego-compensated motion and language remains unoccupied by prior work. We fill it with a plug-in that recovers the motion-class expressions camera-naïve hosts systematically miss, leaving their appearance grounding intact and demanding no retraining of the host it attaches to.

3 METHOD

GMC-Link sits between an RMOT host and its scoring step. It assumes a two-stage, referring-by-tracking host interface — tracked boxes and a per-track, per-expression score to correct — which both hosts we study expose. It reads the host’s tracked boxes, estimates how the camera moved, subtracts that motion from each object’s trajectory, aligns the residual kinematics with the referring expression, and adds a calibrated correction to the host logit. No host weight is retrained and no detector is replaced; each host receives a small per-axis scalar calibration. Figure 1 gives the architecture-level view of this attachment point, and Figure 2 shows the four stages.

The architecture separates three responsibilities that are otherwise entangled in camera-naïve RMOT hosts. The host remains responsible for detection, association, and appearance-language grounding; the GMC branch independently estimates background camera motion and converts the host’s track boxes into residual object motion; the fusion layer then treats the motion-language score as a calibrated correction rather than as a replacement for the host score. This separation is the reason the module can be attached after host inference without retraining the host or changing its detector.

3.1 Ego-motion compensation

The first stage recovers the camera transform between consecutive frames so that object displacement can later be separated from camera displacement. We detect ORB keypoints [12] in both frames and match descriptors with a brute-force Hamming matcher, retaining a correspondence only when it passes Lowe’s ratio test at 0.7. Background dominates the matched set: we seed a foreground mask from the host’s boxes in the previous frame and suppress keypoints inside them, so the surviving matches lie on static scene structure rather than on the moving objects themselves. RANSAC [3] then fits a 3×3 homography $H_{t-1 \rightarrow t}$ to the masked correspondences, rejecting parallax outliers, and we record the median inlier reprojection error as a per-frame background residual. When too few inliers survive, the stage falls back to identity, so a frame with weak texture degrades to “no compensation” rather than to a corrupted estimate.

A single homography is exact only for planar or pure-rotation scenes. Driving footage is planar-dominated, which is why ORB with RANSAC outperformed dense optical flow on this data: the explicit outlier rejection discards the residual parallax that a flow field would smear into the estimate. Reasoning over a window of frames requires composing these transforms. Rather than warp coordinates iteratively, which accumulates interpolation error, we store the original never-warped object centroids and compose a single cumulative homography spanning k frames,

$$H_{t-k \rightarrow t} = H_{t-1 \rightarrow t} H_{t-2 \rightarrow t-1} \cdots H_{t-k \rightarrow t-k+1}, \quad (1)$$

applying it once to map a past centroid into the current frame. One warp per query is numerically more stable than k chained warps.

3.2 Multi-scale residual velocity

Compensated geometry lets us isolate true object motion. For a temporal gap g , the raw object velocity is the displacement of its centroid; the ego velocity is the displacement that the camera alone would induce, obtained by warping the past centroid through Eq. 1. The residual velocity is their difference,

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}}, \quad (2)$$

which is the motion a stationary camera would have observed. A parked car has near-zero residual velocity even while its pixels sweep across the frame, which is exactly the case that camera-naïve hosts misread. We normalise each residual by image dimensions, $v_{\text{norm}} = (v_{\text{pix}}/d_{\text{img}}) \times 100$, so the representation is resolution-invariant and lands near unit magnitude.

We compute Eq. 2 at three frame gaps, $\{2, 5, 10\}$, to capture short, mid, and long-range motion in one descriptor. Concatenating these residuals with bounding-box state yields a 13-dimensional motion vector, $[dx_s, dy_s, dx_m, dy_m, dx_l, dy_l, \Delta w, \Delta h, c_x, c_y, w, h, \text{snr}]$, where the six leading entries are residual (dx, dy) velocity pairs at gaps 2, 5, and 10 rather than all- x followed by all- y . The multi-scale design separates short jitter from sustained translation that a single gap conflates. A gap of 2 resolves frame-to-frame jitter that a gap of 10 averages away, while the gap of 10 accumulates a slow drift that is buried in the noise of a single short step, so the three scales are complementary rather than redundant. The final entry, snr , is a signal-to-noise ratio: the mid-scale residual object speed divided by a background noise floor built from the buffer of RANSAC inlier residuals from Stage 1. It reports how trustworthy the compensation was on that frame; it leaves mean separation unchanged but reduces per-track score variance (a different quantity from the seed-to-seed spread in Table 4) by down-weighting frames where the homography was poorly constrained.

3.3 Motion-language alignment

A lean two-tower encoder maps the 13D motion vector and the expression into a shared space. Each modality passes through a Linear adapter (motion $13 \rightarrow 256$, language $384 \rightarrow 256$, the latter from an all-MiniLM-L6-v2 SentenceTransformer [11]) into a shared MLP ($256 \rightarrow 512 \rightarrow 512 \rightarrow 256$) followed by layer normalisation and ℓ_2 normalisation. We train it with a symmetric InfoNCE objective [13] at temperature 0.07 and mask false negatives that arise when distinct samples share an expression. At inference we read the raw cosine similarity directly, with no sigmoid and no temporal smoothing, so the fusion stage receives an uncalibrated alignment score. The encoder choice is deliberate: swapping the language tower to CLIP-text lifts pooled HOTA on one host but regresses within-class scores (moving -1.03 , static -1.74 ; a trajectory-pooling artifact) and is negative on all three settings under the shipped raw-cosine pipeline, and larger sentence encoders (768D) are flat to negative, so the lean MiniLM tower stays. This aligner is a standard contrastive component; its ceiling is representation-bound, and the design budget is spent on compensation and fusion rather than on the tower.

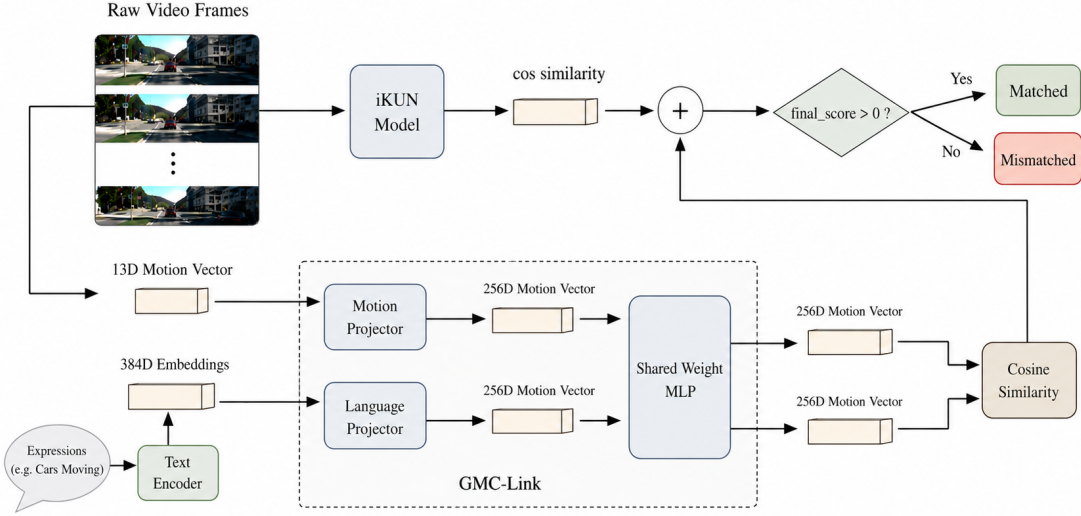


Figure 1: GMC-Link architecture. The RMOT host provides tracked boxes and a host expression score. GMC-Link estimates inter-frame camera motion from the video, derives ego-compensated residual motion from the host tracks, aligns that motion descriptor with the referring expression, and sends only a calibrated score correction back to the host decision layer. The host’s appearance pathway and tracker remain unchanged.

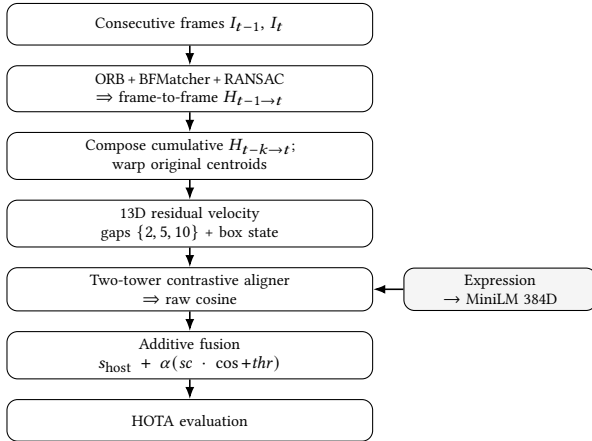


Figure 2: GMC-Link pipeline. Consecutive frames feed an ORB homography estimate; transforms compose into a cumulative homography that yields a 13D residual-velocity descriptor; a two-tower encoder aligns the descriptor with the referring expression; the alignment score corrects the host logit through additive fusion, and the result is scored by HOTA.

3.4 Decision-level additive fusion

The alignment score corrects the host logit at the decision level. We inject it additively per host and per axis,

$$s_{\text{final}} = s_{\text{model}} + \alpha (sc \cdot \cos + thr), \quad (3)$$

where $\cos$ is the raw motion-language cosine, sc rescales it onto the host’s logit range, thr shifts the operating point, and α sets the overall weight. Feature-level injection of motion into the visual encoder was catastrophic in an early trial (−21.7% F1, a historical diagnostic on a prior pipeline), which motivates keeping the correction at the decision layer where it cannot distort the host’s appearance representation.

A keyword router classifies each expression as a motion query or an appearance query and selects the axis-specific recipe: an expression routes to the motion axis when it contains any of a fixed list of 15 motion keywords (substring match; e.g. *moving*, *turning*, *parked*, *braking*) and to the appearance axis otherwise, so a mixed expression such as “moving black cars” takes the motion recipe. The two axes differ sharply in their scale calibration: appearance expressions such as “black cars” carry no motion content, so the GMC score is noise on that axis and is damped by an sc_{appear} that is roughly seven to eleven times smaller than sc_{motion} . The recipes also absorb the host logit scale, which differs by an order of magnitude between the hosts (V1-test score standard deviations 0.50 for iKUN [2] versus 10.2 for FlexHook [6], a 20× spread ratio matched by the 11× scale ratio in Table 1). Table 1 lists the locked coefficients per host-split setting; the same module attaches to both TransRMOT-lineage [14] host architectures we study through this calibration alone.

These coefficients are scalars, not a learned head. We tried to derive them automatically by matching the standard deviations of the two score streams, and we tried replacing them with a learned fusion network; both underperformed the hand-set values, so the

Table 1: Locked fusion coefficients for Eq. 3, by host-split setting and axis. FlexHook (V1) and (V2) are the same architecture calibrated per split. Appearance scale is damped relative to motion scale because the motion signal is uninformative on appearance expressions.

Host	Axis	α	sc	thr
iKUN	motion	1.00	0.90	+0.17
	appearance	1.00	0.30	+0.10
FlexHook (V1)	motion	0.65	10.0	+3.0
	appearance	1.00	3.50	+0.9
FlexHook (V2)	motion	0.40	10.0	+1.3
	appearance	1.00	3.50	+1.2

calibration stays a small fixed table that a new host can fit in a single pass without retraining its tracker.

4 EXPERIMENTS

4.1 Setup

We evaluate on Refer-KITTI, the standard RMOT benchmark, in two splits: Refer-KITTI V1 [14] (3-sequence pooled test) and its expanded relabeling Refer-KITTI V2 [16] (4-sequence pooled test). All scores are HOTA [8] computed with TrackEval. We study two host architectures, iKUN [2] and FlexHook [6], in three host-split settings: FlexHook (V1) and (V2) denote one architecture trained and evaluated per split, not two model variants. Frame numbering must agree between the ground-truth template and the tracker output. We use the paper-canonical convention aligned with the NeuralSORT detections, under which we adopt the public iKUN [2] pipeline’s published 44.564 pooled HOTA as the comparison anchor. The alternative regeneration is off by one frame and drops HOTA by roughly six points, so we do not use it. Every result with our module is reported as a mean over independent training seeds with its standard deviation: $n=3$ (seeds $\{0, 1, 2\}$) for the cross-setting tables, and $n=5$ (seeds $\{0, \dots, 4\}$) for the component ablation and its full-module reference, whose arms share seeds and are compared with Welch’s t -test. The aligner trains with AdamW (learning rate 10^{-3} , batch 256, 100 epochs) under symmetric InfoNCE at temperature 0.07; ORB uses a 1500-feature budget and RANSAC a 5 px reprojection threshold. Code, trained aligner weights for every seed, and the per-sequence score caches will be released upon publication. For each host we report two references: the unmodified host (*native*) and the host fused with raw-cosine GMC scores under a simple, uncalibrated additive rule (*+GMC simple*), which separates the value of the per-host recipe from the value of any GMC signal at all. Because Refer-KITTI provides only three (V1) to four (V2) test sequences and no held-out tuning split, the six per-host coefficients in Table 1 are calibrated on the evaluation data; they are fixed scalars rather than a learned head, but we report them as a per-host calibration to this benchmark rather than a transferable setting. A leave-one-sequence-out check bounds the overfitting risk and validates transfer directly on all three settings: each fold refits the coefficients by the same grid on the remaining test sequences only and scores the held-out sequence frozen.

On iKUN the moving-class recovery survives essentially intact — LOSO moving-class HOTA 28.38 ± 1.60 versus 29.47 ± 0.88 in-sample on the same three seeds, against a native 20.25 — and eight of the nine fold-fits select a motion scale within $[0.7, 1.1]$ of the shipped 0.9; LOSO pooled HOTA is 44.32 ± 0.09 , giving back 0.32 to sit at native parity rather than above it. FlexHook transfers the same way: LOSO pooled 53.28 ± 0.17 (V1) and 42.74 ± 0.07 (V2) against in-sample 53.526 and 42.807, and the V2 figure stays above the published 42.526, so the one outright pooled improvement survives out-of-fold calibration. We accordingly claim the moving-class effect and the V2 pooled gain, not the iKUN pooled number, as the calibration-robust results.

The test splits are heavily skewed toward appearance. V1 holds 158 expressions over 1010 frames, of which 75.3% are appearance queries, 7.6% are static-motion queries, and only 17.1% (27 expressions) refer to a moving object; V2 holds 862 expressions over 1469 frames with a similar 74.1 / 11.7 / 14.2 split. The motion class our module targets is therefore a roughly one-in-six minority of the test set. This shapes how the numbers should be read. Pooled HOTA is computed once over the union of all expressions’ trajectories, so it is dominated by the detection and association counts of the appearance majority; it is not a class-weighted mean of per-class HOTA, and a moving-class gain does not enter the pooled number in proportion to the moving-class expression count. Because the moving class contributes only a small share of the pooled trajectory mass, a recovery concentrated there registers as a small pooled movement even when the per-class effect is large. We therefore treat per-class moving HOTA as the primary metric and report pooled HOTA alongside it as a secondary, whole-benchmark check.

4.2 Main results

Table 2 reports pooled HOTA for the three host-split settings, native and with the shipped module. On iKUN [2] the module reaches 44.634 ± 0.066 , within one standard deviation of the host’s own published 44.564: it matches the pooled score rather than beating it. Naively fused under the uncalibrated rule, the GMC signal is in fact net-negative on two of the three settings (*+GMC simple* 44.272 on iKUN and 53.121 on FlexHook (V1), both below native), so the per-host recipe is not additive polish but a rescaling that turns a mis-scaled signal from harmful to helpful. The motion-class breakdown explains where the effort goes. iKUN reads motion most weakly of the hosts studied, and the module’s per-class pooled gains land squarely on that deficit, with the largest single per-class delta of +8.83 on the iKUN moving class (Table 3). Across the nine setting-by-class per-class pooled deltas (three settings $\times$ motion/static/appearance), all nine are positive relative to the native host under $n=3$ evaluation, each delta being HOTA recomputed on that class’s own trajectory subset. Pooled HOTA is not additive over these subsets — its association term couples every identity in the union — so per-class gains need not sum to a pooled gain: on iKUN the recoveries net to pooled parity, and on FlexHook (V1) they coexist with a residual pooled shortfall against the host’s published figure (below). On the V2 split, FlexHook shows the clean pooled improvement, reaching 42.807 ± 0.038 against the

Table 2: Pooled HOTA on Refer-KITTI test, native host versus the shipped GMC module ($n=3$, mean $\pm$ std). Δ is against each host's published pooled HOTA. iKUN *matches* its paper score while lifting the motion class (Table 3); the iKUN Δ of +0.070 lies within the ± 0.066 seed spread and is not a beat. FlexHook (V2) is the clean pooled improvement. The *+GMC simple* control column is the simple-recipe anchor ($\alpha=1$, $sc=1$, $thr=0$, raw cosine, sw aligner, $n=3$).

Host	Native	+GMC simple	+GMC (ship)	Δ paper
iKUN [2] (V1)	44.564	44.272	44.634 $\pm$ 0.066	+0.070
FlexHook (V1) [6]	53.824	53.121	53.526 $\pm$ 0.087	-0.298 [†]
FlexHook (V2) [6]	42.526	42.532	42.807 $\pm$ 0.038	+0.281

[†]FlexHook (V1) paper gap is structural (no tested configuration beats 53.824); Δ vs *+GMC simple* anchor = +0.405.

published 42.526, a gain of +0.281 that survives the appearance dilution. On V1, FlexHook improves over the *+GMC simple* anchor (53.121 $\rightarrow$ 53.526) but retains a residual gap to its own published number.¹ For external context, the strongest motion-language host on this benchmark, VMRMOT [9] (53.00/35.21 HOTA on V1/V2), uses its own end-to-end detector and tracker, so a direct row is confounded by pipeline rather than by our module; we accordingly compare each host against its own native score rather than across pipelines.

4.3 The motion deficit

The size of the gain is not uniform across settings; it tracks how much motion ability the host lacks in that setting to begin with. Table 3 pairs each setting's native moving-class HOTA with the moving-class gain the module adds. iKUN starts at the lowest native moving-class ability and gains the most (+8.83 $\pm$ 0.83, $n=5$); FlexHook on V1 starts higher and gains little (+0.91 $\pm$ 0.79, $n=3$); on V2 it starts highest and gains almost nothing (+0.43 $\pm$ 0.17, $n=3$). The relationship across the three settings is monotone-inverse: a setting with a larger motion deficit receives a larger motion-class correction. We do not state this as a proportional law; three settings cannot support one. What it does support is a prediction, and the two FlexHook settings are the predicted low-deficit cases where a small deficit yields a small gain, consistent with the module supplying motion evidence only where the host fails to.

4.4 Ablations

To locate where the recovery comes from, we strip the module one component at a time and read the moving-class HOTA, the metric the pooled score hides, over $n=5$ seeds per arm. Table 4 reports the full module against three reductions: replacing residual velocity with raw, uncompensated velocity (*-ego*); collapsing the three temporal gaps to one (*-multiscale*); and removing the signal-to-noise feature (*-snr*). The recovery is carried first by the module itself:

¹FlexHook's published V1 pooled HOTA is 53.824, which neither the shipped module (53.526, $\Delta = -0.298$) nor any tested configuration beats — the strongest, a legacy per-modality-MLP aligner with temporal smoothing, reaches 53.716 $\pm$ 0.068 ($n=3$) and still trails; the gap is structural rather than a tuning artifact, and we treat the V1 paper comparison as out of scope under the FlexHook split convention while reporting the gain against the *+GMC simple* anchor.

Table 3: Motion deficit across host-split settings. Native moving-class HOTA versus the moving-class HOTA gain added by the module. The gain is monotone-inverse to native motion ability across the three settings.

Host	Native MOVING	GMC MOVING gain
iKUN [2]	20.25	+8.83 $\pm$ 0.83 ($n=5$)
FlexHook (V1) [6]	43.98	+0.91 $\pm$ 0.79 ($n=3$)
FlexHook (V2) [6]	48.02	+0.43 $\pm$ 0.17 ($n=3$)

Table 4: Component ablation in moving-class HOTA (the metric pooled scores dilute), iKUN host, $n=5$ seeds (mean $\pm$ std). Against the native host (20.25), the full module recovers +8.83; ego-compensation is the only component whose removal costs a significant amount (-1.77 , Welch $p=0.006$), while collapsing the temporal scales or removing snr fall within the seed spread.

Variant	MOVING HOTA
Native host (no module)	20.25
Full module	29.08 $\pm$ 0.83
–ego (raw velocity)	27.31 $\pm$ 0.66
–multiscale (single gap)	28.82 $\pm$ 0.84
–snr	29.74 $\pm$ 0.59

aligning motion with language lifts the moving class from 20.25 to 29.08 (+8.83 over the native host). Among the engineered components, only ego-compensation shows a significant effect: feeding raw velocity, which carries the camera's motion intact, costs -1.77 (Welch $t=3.75$, $p=0.006$) and gives back a fifth of the module's gain, indicating that compensation contributes over and above merely having a motion descriptor. Collapsing the temporal scales (-0.26 , $p=0.64$) and removing snr ($+0.67$, $p=0.19$) both fall inside the seed spread, so the multi-scale and snr designs earn their place as resolution and robustness choices rather than as measured moving-class gains.

The injection site is as consequential as the signal. Injecting motion features into the visual encoder, rather than fusing at the decision level, regressed the host by -21.7% F1 in that early feature-level trial, which is why the module corrects the logit after the host has scored appearance rather than reshaping the host's representation.

The GMC module runs at roughly 68 FPS, a median 14.8 ms per frame end-to-end (timed over the ORB homography, descriptor, and alignment stages, excluding the host's own inference), of which the ORB homography on CPU is the bottleneck at 14.55 ms while the aligner forward is under 0.2 ms — about $6.7\times$ the benchmark's 10 fps capture rate, so the compensation adds no practical inference burden.²

²A separate appearance re-ranker, orthogonal to this paper's motion contribution and targeting the appearance majority rather than the motion class, stacks on iKUN to 45.612 pooled HOTA ($n=3$); we note it only to bound what the motion module alone contributes and do not claim it here.



Figure 3: Real recovery under camera ego-motion on “moving cars” (Refer-KITTI seq 0005, frames 85/94/100 of a forward-driving clip; the green car moves, the red car is parked). The dashed line marks the mover’s image column. **Red:** the parked car’s centroid slides left as the camera passes it (center- x 261 $\rightarrow$ 104 px), so the camera-naive host scores it “moving” (false positive), yet its ego-compensated residual is near zero and the module restores it to static. **Green:** the moving car holds its column (center- x $\approx$ 605, < 1 px/frame), so the host misses it (false negative), but its nonzero residual — unlike the static background — recovers it as moving. Both errors flip from the same host logit through the additive correction.

5 CONCLUSION AND LIMITATIONS

A plug-in decision-level module that compensates camera ego-motion recovers the motion-class referring expressions that camera-naive hosts systematically miss. The recovered signal grows as the host’s own ability to read object motion shrinks: across the three host-split settings studied, the gain is monotone-inverse to native motion ability. The module delivers this at a fixed tracker, retraining neither the host nor swapping the detector, and adds only a small per-setting scalar calibration. On iKUN [2] it matches the published pooled HOTA while lifting the motion class the pooled number hides; on FlexHook [6], on the V2 split, it improves the pooled score outright.

We frame the open issues as bounded failure modes. The compensation rests on a single homography per frame, estimated by the same ORB-and-RANSAC pipeline that plain-tracking GMC [1] uses — so the front-end is a mature component we reuse rather than a new estimator to baseline against — and is exact only for planar or pure-rotation scenes. Driving is planar-dominated, and the parallax that survives the ORB-and-RANSAC fit is discarded as outliers and further absorbed by an aligner trained on jittered,

imperfect motion, bounding strong parallax at two stages. The foreground mask is seeded from the host’s own boxes, so a badly wrong detection can leak object features into the ego estimate; RANSAC and an identity-homography fallback bound that leakage, so compensation degrades gracefully with detection quality. Our evidence is confined to Refer-KITTI driving footage, where the planar-dominated assumption holds; we do not claim transfer to non-planar or non-driving moving-camera video, such as handheld or aerial capture, where a single homography no longer models the scene. Our evidence also covers only two-stage, referring-by-tracking hosts; end-to-end hosts expose an analogous per-query score, but attachment there is untested. Finally, hosts with native temporal memory, such as TempRMOT [16], fall outside our target: their recurrent state already aggregates trajectory history, making an external motion module redundant rather than complementary.

The open lever is a stronger motion representation: the alignment ceiling sits in the descriptor, not the fusion stage, so richer features encoding the trajectory geometry a 2D vector flattens are the natural next step for the TransRMOT [14] lineage of camera-naive hosts.

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